1) $\frac{d^4y}{dx^4} - 13\frac{d^2y}{dx^2} + 36y = 0$

$r^4 - 13r^2 + 36 = 0$

$x_1 = 3 \quad x_2 = 2 \quad x_3 = -2 \quad x_4 = -3$

$y = c_1 e^{3x} + c_2 e^{2x} + c_3 e^{-2x} + c_4 e^{-3x}$

2) $\frac{d^3y}{dx^3} + 3\frac{d^2y}{dx^2} - 4\frac{dy}{dx} - 12y = 0$

$r^3 + 3r^2 - 4r - 12 = 0$

$x_1 = 2 \quad x_2 = -2 \quad x_3 = -3$

$y = c_1 e^{2x} + c_2 e^{-2x} + c_2 e^{-3x}$

3) $\frac{d^2y}{dx^2} - 7\frac{dy}{dx} + 12y = 0$

$r^2 - 7r + 12 = 0$

$x_1 = 4 \quad x_2 = 3$

$y = c_1 e^{3x} + c_2 e^{4x}$

4) $\frac{d^2y}{dx^2} + y = 0$

$r^2 + 1 = 0$

$x_1 = i \quad x_2 = -i$

$y = c_1 \cos i + c_2 \sin x$

5) $\frac{d^4y}{dx^4} - y = 0$

$r^4 - 1 = 0$

$x_1 = 1 \quad x_2 = -1 \quad x_3 = i \quad x_4 = -i$

$y = c_1 e^{x} + c_2 e^{-x} + c_3 \cos x + c_4 \sin x$

6) $\frac{d^3y}{dx^3} - 5\frac{d^2y}{dx^2} + 7\frac{dy}{dx} = 0$

$r^3 - 5r^2 + 7r = 0 \Rightarrow \quad A = \frac{5}{2} \quad B = \frac{\sqrt{3}}{2}$

$x_1 = \frac{5 + \sqrt{3}i}{2} \quad x_2 = \frac{5 - \sqrt{3}i}{2} \quad x_3 = 0$

$y = c_1 + e^{\frac{5}{2}x}\left(\frac{5}{2} c_2 \cos\left(\frac{\sqrt{3}}{2}x\right) + \frac{5}{2} c_3 \sin\left(\frac{\sqrt{3}}{2}x\right)\right)$

7) $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 0$

$r^2 - 4r + 4 = 0 \quad x_1 = 2$

$$y = c_1 e^{2x} + c_2 x e^{2x}$$

8) $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 0$

$r^2 + 2r + 1 = 0$

$x_1 = -1$

$$y = c_1 e^{-x} + c_2 x e^{-x}$$

9) $\frac{d^4y}{dx^4} + \frac{d^3y}{dx^3} - 3\frac{d^2y}{dx^2} - 5\frac{dy}{dx} - 2y = 0$

$r^4 + r^3 - 3r^2 - 5r - 2 = 0$

$x_1 = 2 \qquad x_2 = -1$

$$y = c_1 e^{2x} + c_2 e^{-x} + c_3 x e^{-x} + c_4 x^2 e^{-x}$$